

Nieb Hasan Neom

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Education

PhD, Computer Science, Missouri University of Science and Technology Kummer Innovation and Entrepreneurship Doctoral Fellow · Expected May 2030	2025 – 2030
BSc, Computer Science and Engineering, Khulna University of Engineering & Technology KUET Technical Scholarship, 2017–2022	2017 – 2022

Research

Graduate Research Assistant, Computer Systems & Networking Lab, Missouri S&T AI systems for data movement and scientific workflows, advised by Dr. Md Arifuzzaman. Transfer systems are evaluated on the CloudLab, FABRIC, and TACC/ACCESS testbeds.	2025 – now
– NetBench: domain adaptation for networking LLMs (Released v1.1.0 · paper in preparation). Benchmark and controlled study of LLM domain adaptation for high-performance networking: 242 questions from 175 papers, 8 open-weight models (1B–27B) under fine-tuning, continued pre-training, and retrieval, two LLM judges validated by a human study, and one-command reproduction of every number. – Guarded orchestration for scientific models (Accepted · WORKS26 at SC26). State-aware layer over LLM function calling that accumulates inputs across turns, checks arguments against typed schemas, and blocks execution until the required state is complete; instantiated for protein-corona prediction. – Agentic Data Transfer Optimizer (In progress · prototype built, evaluation under way). Diagnostic and Decision agents tune transfer concurrency from live TCP telemetry over zero-copy sendfile() workers, inside hard limits with bounded changes and oscillation damping.	

Industry

Software Engineer II, Kona Software Lab Dhaka, Bangladesh · Joined as a Graduate Engineer and was promoted in 2024.	2022 – 2025
– Owned 6 of the 30 Spring Boot microservices behind Kona Token Trade and contributed to 15, across trade, wallet, user, payment, and fee management. – Replaced synchronous REST calls between services with Kafka messaging, so one unavailable downstream service could no longer fail an in-flight transaction. – Added two-factor authentication to buy and sell operations and integrated bank-account and Kona card payment gateways. – Built the Card Personalization System against GlobalPlatform, Visa, Mastercard, and EMV specifications. It is in commercial use, and I was chosen for a 14-day training program on it at parent company Kona I in Seoul. – Mentored and onboarded two junior engineers.	

Publications

N. H. Neom, M. R. Gartia, M. Arifuzzaman. Guarded, State-Aware Orchestration for Schema-Constrained Scientific Model Execution. 21st Workshop on Workflows in Support of Large-Scale Science (WORKS26), held with SC26, 2026 (accepted).	
R. M. Swargo, N. H. Neom, M. Arifuzzaman. Self-Tuning Segmented Retrieval of Large-Scale Genomic Data. 22nd IEEE International Conference on e-Science (eScience 2026), short paper, 2026 (accepted).	
M. A. Habib, M. M. Rahman, N. H. Neom. CredSec: A Blockchain-based Secure Credential Management System for University Adoption. arXiv preprint 2406.05151, 2024 (preprint).	
S. Das, M. S. Imtiaz, N. H. Neom, N. Siddique, H. Wang. A hybrid approach for Bangla sign language recognition using deep transfer learning model with random forest classifier. Expert Systems with Applications, vol. 213, article 118914, 2023 (published).	
M. M. Islam, N. Neom, M. S. Imtiaz, S. Nooruddin, M. R. Islam, M. R. Islam. A Review on Fall Detection Systems Using Data from Smartphone Sensors. Ingénierie des Systèmes d'Information, vol. 24, no. 6, pp. 569–576, 2019 (published).	
Software: N. H. Neom, R. M. Swargo, M. Arifuzzaman. NetBench v1.1.0. Zenodo, 2026. doi:10.5281/zenodo.21892017	

Skills

Languages: Python, Java, C, C++, SQL, Bash AI systems: LLM function calling and tool use, Schema-constrained generation, Agentic workflow orchestration, Ollama and Hugging Face models, PyTorch, Pydantic, NumPy	Systems and networking: TCP performance analysis, Concurrency tuning and multiprocessing, Zero-copy sendfile(), Linux systems programming, CloudLab, FABRIC, TACC/ACCESS Backend: Spring Boot, Apache Kafka, Microservices and REST APIs, AWS S3, SNS, EC2, PostgreSQL, MongoDB, Redis, MySQL Tools: Git, Gradle, Jenkins, Jira, LaTeX
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Honors and service

Honors: Codeforces Expert (peak rating 1629); LeetCode Knight (peak contest rating 1895); First runner-up, KonaSL Coding Combat 2023 (20 teams); KUET Technical Scholarship (2017–2022).	
Service: Student volunteer, SC26; Reviewer, Journal of Autonomous Intelligence; Competitive-programming workshops at KUET.	